

BHUMI CHOUDHARY

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Education

University of Illinois Chicago (UIC) – Chicago, IL

Bachelor of Science in Data Science, Concentration in Bioinformatics | Expected: May 2028

- Dean's List, College of Engineering
- Relevant Coursework: Program Design I & II, Foundations of Computing, Data Structures (*IP*), Data + Artificial Intelligence, Probability & Statistics (*IP*), Calculus I–III (*III IP*), Genetics (*IP*)

Hansraj Model School (CBSE Board) – India

Science Stream (Physics, Chemistry, Biology, Math) | Graduated: May 2023

Research Experience

Participant — Argonne National Laboratory HPC Bootcamp — August 2025

- Explored training, fine-tuning, and evaluation of large language models (LLMs) for scientific applications, focusing on model behavior, limitations, and scaling concepts
- Collaborated with peers and gained hands-on exposure to parallel training workflows in Python, applying multi-GPU concepts in a high-performance computing environment

Research Assistant, 3K Nano LLP (Incubated at JNU) — Jan 2024 – July 2024

- Contributed to the “Aptamer-based Lateral Flow Assay (LFA)” project
- Assisted in experimental design, data collection, analysis for diagnostic assay development

Researcher, JNU (Review Article Project) — June 2023 – Jan 2024

- Conducted a systematic review of 80+ peer-reviewed studies on nanosponges for oil spill remediation and wastewater treatment
- Synthesized comparative insights on materials properties, scalability, and environmental impact into a review manuscript

Research Volunteer, Special Centre for Nanoscience, JNU — Summer 2022

- Conducted nanofiber synthesis and antibacterial testing under faculty supervision
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Leadership Experience

Teaching Assistant, CS111 (Program Design) — Fall 2025

- Supported course instruction for an introductory Python programming course by assisting the professor and instructional team during lectures and labs
- Guided 30+ students in weekly lab sessions and in-person office hours by debugging programs, reviewing assignments and projects, resolving logic and runtime errors, and reinforcing effective coding and problem-solving practices

Graphic Designer — WeDesi (UIC South Asian Student Organization) — Fall 2025

- Led graphic design for digital and print marketing across multiple student events, ensuring consistent branding and clear event communication
- Served as a core board member, collaborating on event planning and managing logistics including decorations, catering coordination, timelines, and on-site execution

Representative — Undergraduate Student Government (USG), Student Success Committee — Fall 2025

- Represented student perspectives on academic and student-success initiatives; volunteered in USG events supporting outreach and engagement
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Volunteer and Outreach

Rotaract Club of Rever (2023–2024)

- Organized weekly community service projects (~4 hrs/week), supporting social causes

Mentor (2020–2022)

- Tutored under-resourced students in science and math
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